

Stage 1: Raw Data Ingestion

NYC Yellow Taxi Trip
Records, GPS
coordinates,
Timestamps

Stage 2: Preprocessing

Outlier removal,
Temporal binning,
Spatial clustering K=40
MiniBatch KMeans

Stage 3: Feature Engineering

Lag features, Temporal
cyclicals, Weather
features

Stage 4: Model Training

8 individual models, 6
ensemble strategies

Stage 5: Evaluation

RMSE, MAE, MAPE, R^2 ,
RMSLE, Friedman test

Stage 6: Ride Pooling + Impact

Multi-criteria greedy
matcher, Bootstrap CI,
Cost/ CO_2 savings